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August 11, 2025

**Re: Manuscript submission, Submission ID 409d7839-7d85-411e-ad16-75038b1eacad**

Dear Dr. Zandomeneghi,

I am sending you our manuscript entitled “Genestrip: Space-efficient  $k$ -mer counting, read classification and filtering for selected groups of organisms” by Daniel Pfeifer et al. (Submission ID 409d7839-7d85-411e-ad16-75038b1eacad). We would like to have the manuscript considered as a software article in BMC Bioinformatics. In our manuscript we present and evaluate Genestrip as a new type of bioinformatics tool for  $k$ -mer based metagenomics. With our contribution, we hope to considerably advance the efficiency of metagenomic data analysis.

The consumption of main memory resources is a significant burden in  $k$ -mer-based metagenomic analysis when creating related databases but also when performing (unique)  $k$ -mer-counting and read classification. Genestrip addresses this issue by focusing on small but freely configurable groups of organisms. Regarding the selected organisms, it produces  $k$ -mer databases and results comparable to those of KrakenUniq but at a fraction of its required memory resources. Our tool follows a new strategy to compute the lowest common ancestor taxon for a stored  $k$ -mer which enables read analysis with high accuracy for the organisms of interest.

We assess the correctness, usefulness and performance of Genestrip in different contexts and show that it indeed ascertains high accuracy for organisms whose genomes were included in a corresponding database. Our example databases comprise a dozen to a few thousands of species and lend themselves to usage in tick surveillance, medical diagnostics or agriculture. The focus on a certain set of organisms allows for more genomes to be included from the organisms while the resulting database remains small. Such small but deep databases tend to improve recall during read classification while sustaining high precision.

On the whole, Genestrip's qualities empower bioinformaticians and other users to flexibly design, build and use small *k*-mer databases for their own needs with potentially deep genomic coverage.

We confirm that our manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the manuscript and agree with its submission to BMC Bioinformatics. Daniel Pfeifer (myself) is the corresponding author.

Please let me know of your decision at your earliest convenience.

With my best regards,



**Daniel  
Pfeifer**

Daniel Pfeifer

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